

J. Jake Nichol

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Research Interests

My research advances AI Credibility for high-consequence scientific and national security applications: I believe consequential decisions demand mechanistic understanding, not black-box correlation. I develop causal discovery and physics-informed modeling methods to improve the interpretability, robustness, and evidence-based assurance of AI-enabled systems, including foundation and generative models. This is motivated by a broader interest in discovering new dynamics and phenomena in high-consequence domains and the natural sciences, particularly Earth systems, and in how credibility evidence supports human trust and decision-making.

Education

- 2018 – 2025 **Ph.D. Computer Science, University of New Mexico**, School of Engineering.
Moses Biological Computation Lab.
Dissertation: *Seeking Structure in Complex Systems: From Feature Analysis to Space-Time Causal Discovery with Earth Science Applications*
- 2015 – 2017 **M.B.A., University of New Mexico**, Anderson School of Management.
- 2011 – 2016 **B.S. Computer Science, University of New Mexico**, School of Engineering.

Employment History

- 2025 – . . . **Postdoctoral Appointee**, Scientific Machine Learning, Sandia National Laboratories.
- 2021 – 2025 **R&D Graduate Intern (Year-Round), Full-Time**, Scientific Machine Learning, Sandia National Laboratories.
- 2019 – 2021 **R&D Graduate Intern (Year-Round), Part-Time**, Scientific Machine Learning, Sandia National Laboratories.
- 2018 – 2019 **Graduate Research Assistant**, Department of Computer Science, School of Engineering, University of New Mexico.
- 2017 – 2019 **Owner**, Swarming Technologies LLC: profitable robotics company selling self-designed autonomous robots, custom-design consulting, and repair services.
- 2015 – 2017 **Robot Engineer/Designer**, NASA Swarmathon & Moses Biological Computation Lab, Department of Computer Science, School of Engineering, University of New Mexico.
- 2014 – 2014 **Software Engineering Intern**, Intel Corporation.

Research Publications

Journal Articles

- 1 **J. J. Nichol**, M. Weylandt, G. M. Fricke, M. E. Moses, D. Bull, and L. P. Swiler, “Space-Time Causal Discovery in Earth System Science: A Local Stencil Learning Approach,” *Journal of Geophysical Research: Machine Learning and Computation*, vol. 2, no. 3, 2025, ISSN: 2993-5210. 🌐 DOI: 10.1029/2024jh000546.
- 2 **J. J. Nichol**, M. G. Peterson, K. J. Peterson, G. M. Fricke, and M. E. Moses, “Machine Learning Feature Analysis Illuminates Disparity Between E3SM Climate Models and Observed Climate Change,” *Journal of Computational and Applied Mathematics*, vol. 395, p. 113 451, Oct. 2021, ISSN: 0377-0427. 🌐 DOI: 10.1016/j.cam.2021.113451.

Accepted / In Press

- 1 E. B. Adikaibe, A. I. Dessanti, **J. J. Nichol**, M. A. Smith, and M. C. Darling, “A Multi-Objective Optimization Framework for Trustworthy Machine Learning,” in *Proceedings of Modeling Decisions for Artificial Intelligence (MDAI) 2026*, To appear, 2026.
- 2 G. Fragkos, **J. J. Nichol**, T. Y. Yen, L. Blakely, S. Hossain-McKenzie, A. Chavez, and S. Wright, “Causal AI-Powered Dynamic ML Model Orchestration for Robust Cybersecurity,” in *Proceedings of the IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm) 2026*, To appear, 2026.
- 3 **J. J. Nichol**, R. M. Milewicz, A. C. Kinney, S. A. Grayson, M. A. Smith, K. M. Butler, E. C. S. Acquesta, and M. C. Darling, “Challenges of Certifying Machine Learning Trustworthiness in High-Consequence Domains,” *Data Science in Science*, 2026, Accepted.

Workshop Papers

- 1 **J. J. Nichol**, M. Peterson, G. M. Fricke, and K. Peterson, “Learning Why: Data-Driven Causal Evaluations of Climate Models,” in *ICML 2021 Workshop: Tackling Climate Change with Machine Learning*, 2021.  DOI: 10.2172/1884401.
- 2 S. M. Ackerman, G. M. Fricke, J. P. Hecker, K. M. Hamed, S. R. Fowler, A. D. Griego, J. C. Jones, **J. J. Nichol**, K. W. Leucht, and M. E. Moses, “The Swarmathon: An Autonomous Swarm Robotics Competition,” in *ICRA 2018 Workshop: Swarms — From Biology to Robotics and Back*, 2018.  DOI: 10.48550/arxiv.1805.08320. eprint: 1805.08320.

Under Review

- 1 **J. J. Nichol**, L. Blakely, and G. Fragkos, “Causal Graphs for Threat Detection in Cyber-Physical Power Grids,” Submitted to IEEE Resilience Week (RWS) 2026, 2026.
- 2 J. S. Pérez-Carrasquilla, **J. J. Nichol**, V. Robledo, D. Bull, K. Dagon, M. N. Evans, and M. J. Molina, “Tracing the Space-Time Causal Origins of Earth System Extremes,” Submitted to Science Advances. Preprint: <https://arxiv.org/abs/2607.10033>, 2026.

In Preparation

- 1 A. R. Croker, D. Guariso, E. Landgren, K. Posch, A. Wiechman, E. Zajdela, and **J. J. Nichol**, “(Opinion) Designing Institutions for an Era of Contested Expertise,” In preparation, target: Proceedings of the National Academy of Sciences (PNAS).
- 2 K. Goode and **J. J. Nichol**, “If All Explanations Are Wrong, but Some Are Useful, How Do We Determine Which Are Credible?” In preparation.
- 3 **J. J. Nichol**, M. Weylandt, G. M. Fricke, and E. M. Moses, “M-CaStLe: Uncovering Local Causal Structures in Multivariate Space-Time Gridded Data,” In preparation, target: Transactions on Machine Learning Research (TMLR). Preprint: <https://arxiv.org/abs/2605.00398>.

Technical Reports

- 1 D. Bull, K. Peterson, L. Shand, L. Swiler, I. Tezaur, B. K. Cook, A. Salinger, C. Amann, B. Watts, R. Leland, L. Bertagna, H. Brown, M. Brown, M. Campos, M. Carlson, K. Chowdhary, J. Crockett, W. Davis, T. Ehrmann, R. Garrett, K. Goode, M. Gulian, C. Hall, G. Harper, J. Hart, J. Hickey, B. Hillman, B. Houchens, J. G. Huerta, D. Krofcheck, J. Li, I. Manickam, K. McClernon, A. McCombs, **J. J. Nichol**, M. Peterson, D. Ries, M. A. Smith, A. Staid, A. Steyer, J. D. Tucker, B. Wagman, J. Watkins, C. Wentland, E. Wenzel, R. M. Weylandt, and A. Yarger, “CLimate Impact: Determining Etiology thRough pAthways (CLDERA),” Sandia National Laboratories, Albuquerque, NM, Tech. Rep. SAND2024-13423R, Sep. 2024.  DOI: 10.2172/2480139.

- 2 **J. J. Nichol**, M. Weylandt, M. Smith, and L. Swiler, “Benchmarking the PCMCI Causal Discovery Algorithm for Spatiotemporal Systems,” Tech. Rep., 2023.  URL: <https://www.osti.gov/biblio/1991387>.
- 3 **J. J. Nichol**, M. Peterson, and K. Peterson, “Causal Evaluations for Identifying Differences between Observations and Earth System Models,” Tech. Rep., 2021.  DOI: 10.2172/1820528.
- 4 K. J. Peterson, A. J. Powell, I. K. Tezaur, E. L. Roesler, **J. Nichol**, M. G. Peterson, W. L. Davis, J. D. Jakeman, D. J. Stracuzzi, and D. L. Bull, “Arctic Tipping Points Triggering Global Change (LDRD Final Report),” Tech. Rep., Sep. 2020.  DOI: 10.2172/1669210.

White Papers

- 1 K. Peterson, W. Davis, M. Peterson, **J. J. Nichol**, K. Chowdhary, and M. D’Elia, “Advancing Sea Ice Predictability in E3SM with Machine Learning,” Tech. Rep., Apr. 2021.  DOI: 10.2172/1769655.
- 2 A. Powell, E. C. S. Acquesta, W. L. Davis, **J. J. Nichol**, I. Tezaur, K. Peterson, S. Rempe, and J. G. Huerta, “Water Cycle-Driven Infectious Diseases as Multiscale, Reliable, Continuously Updating Water Cycle Sensors,” Tech. Rep., Feb. 2021.  DOI: 10.2172/1769797.

Funded Sandia Projects

FY26 – . . .	Co-Lead, Causal AI Team, MetaShield LDRD (PI: Georgios Fragkos). Combining causal AI with meta-learning, MetaShield is enhancing ML models for detecting cyber-physical threads to power grids. I am co-leading the causal AI team to integrate causal modeling paradigms with ML paradigms for better power grid analysis and resilience. Cross-lab team includes centers 1400, 5600, 8700, and 8800, with industry partners PNM and GE Vernova. Contributor, SEA-CROGS (Scalable, Efficient and Accelerated Causal Reasoning Operators, Graphs and Spikes for Earth and Embedded Systems) (PI: Eric Cyr). Developing causal analyses of the Aurora weather foundation model to determine if systematic interventions can bias model behavior to generate more realistic (or unrealistic) weather phenomena. This project is a collaboration between PNNL and SNL, with academic partners at Brown University, Yale University, California Institute of Technology, New Jersey Institute of Technology, and Stanford University.
FY25 – . . .	Team Lead, Credibility, Towards Certifying Trustworthy Machine Learning LDRD (PI: Michael Darling). Co-leading the credibility assessment effort. Developing the Generalized Modeling Maturity Model (GeMMM), which generalizes the CompSim Predictive Capability Maturity Model (PCMM) to any modeling paradigm. Developing the taxonomy of computational modeling. Cross-lab team includes centers 1400, 5400, 5500, and 8700.
FY22 – FY24	Causal Modeling Lead, Attribution Thrust, CLDERA: CLimate Impact–Determining Etiology thRough pAthways Grand Challenge LDRD (PI: Diana Bull, Thrust Lead: Laura P. Swiler). Led the R&D of applying causal modeling to understand the drivers of stratospheric aerosol transport and their impacts. Led the R&D of a novel causal discovery algorithm for gridded spatiotemporal data in physical systems.
FY21 – FY21	Contributor, Causal Evaluations for Identifying Differences Between Observations and Earth System Models Late-Start LDRD (PI: Matt Peterson). Developed a causal discovery methodology to model Arctic sea ice extent drivers in preparation of applying the workflow in the CLDERA Grand Challenge LDRD.
FY20 – FY22	Contributor, Advanced Data Analytics for Proliferation Detection (ADAPD), a multi-lab initiative. Developed hidden Markov models for multimodal temporal data analysis.
FY18 – FY20	Contributor, Arctic Tipping Points Triggering Global Change LDRD (PI: Kara Peterson). Developed random forest regression models for predicting Arctic sea ice extent with feature importance analysis.

Professional Activities

Conference Talks

- 2025 18th U.S. National Congress on Computational Mechanics (USNCCM18). *Advancing Climate Causal Discovery: Local Space-Time Stencil Learning for High-Dimensional Data*. Chicago, Illinois, United States.
- 2023 American Causal Inference Conference. *Climate Dynamics via Spatially Informed Causal Discovery*. Austin, Texas, United States.
- 2022 European Seminar on Computing (ESCO). *Global Multivariate Causal Discovery for the Analysis of Emergent Properties in Earth System Models*. Pilsen, Czech Republic.
- 2021 Chesapeake Large-Scale Analytics Conference (CLSAC). *Causal Discovery for Climate Science and the Energy Exascale Earth System Model*. Virtual.
- 2020 European Seminar on Computing (ESCO). *Machine Learning to Compare Arctic Simulations with Observed Data*. Pilsen, Czech Republic.

Poster Presentations

- 2025 "Toward Certifying Trustworthy Machine Learning" Darling, M., **Nichol, J. J.**, Acquesta, E., Butler, K., Kinney, A., Milewicz, R., Smith, M., & Speed, A. (2025, February 25-28). Conference on Data Analysis (CoDA). Santa Fe, New Mexico, United States.
- 2024 "Recovering the Spatial Evolution of Volcanic Aerosols." **Nichol, J. J.** (2024, May 7). Sandia National Laboratories and Los Alamos National Laboratories Annual Climate Summit. Los Alamos, New Mexico, United States.
"Recovering the Spatial Evolution of Volcanic Aerosols." **Nichol, J. J.** (2024, April 22). Sandia National Laboratories Analytics for Climate and Earth Sciences (ACES) Symposium. Albuquerque, New Mexico, United States.
- 2021 "Learning Why: Data-Driven Causal Evaluations of Climate Models." **Nichol, J. J.**, Peterson, M. G., Fricke, G. M., & Peterson, K. J. (2021, July 18-24). International Conference on Machine Learning (ICML). Virtual.
- 2020 "Comparing Simulated and Observed Data with Random Forest Feature Importance." **Nichol, J. J.**, Peterson, M., Peterson, K., & Stracuzzi, D. (2020, February 25-27). Conference on Data Analysis (CoDA). Santa Fe, New Mexico, United States.
- 2019 "Feature Comparison of Arctic Observations and Climate Models." **Nichol, J. J.**, Peterson, M., Peterson, K., Stracuzzi, D. Fricke, M., & Moses, M. (2019, December 9-13). American Geophysical Union (AGU) Fall Meeting. San Francisco, California, United States.

Workshops and Invited Participation

- 2025 Santa Fe Institute Workshop (Invited): *Postdocs in Complexity Global Conference: A Postdoc Tapestry, Weaving Global Collaboration*. September 15-18, 2025. Santa Fe, New Mexico, United States.
- 2020 ICML Workshop: *Tackling Climate Change with Machine Learning*. July 23, 2020. Virtual.

Honors and Awards

- 2020 **Best Talk Prize**, European Seminar on Computing (ESCO) 2020
3rd Place Poster Prize, Department of Energy Conference on Data Analysis (CoDA) 2020

Professional Service

- 2026 Reviewer for *IEEE Transactions on Neural Networks and Learning Systems*.
2025 Reviewer for American Meteorological Society's *Journal of Climate*.
2024 Reviewer for *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.

Professional Memberships

- 2025-2027 U.S. Association for Computational Mechanics (USACM)
2023-2025 Society for Causal Inference
2019-2025 American Geophysical Union

Software Packages

Published

- `clif`: CLImate Fingerprinting. *A python library that calculates empirical orthogonal functions for mainly climate data.* Developed with Kenny Chowdhary. Available on  <https://github.com/sandialabs/clif>
- `CaStLe`: Causal Space-Time Stencil Learning. *Local causal space-time structures from observational data.* Available on  <https://github.com/jjakenichol/CaStLe>.

Technical Skills

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|-------------------------|---|
| Causal Inference | Algorithmic causal inference, i.e., causal structure learning, causal discovery, or causal network learning; causal inference techniques for advancing scientific machine learning. |
| Machine Learning | Scientific ML; domain/physics-informed ML; ML feature importance, such as random forests Gini importance, permutation importance, drop-column importance, SHAP. |
| Artificial Intelligence | AI for Earth systems science; trusted AI, explainable AI, and fairness and ethics in AI. |
| Programming | Python and data science libraries such as NumPy, SciPy, Pandas, Xarray, DASK, and Tigramite. |
| Misc. Tech. Skills | L ^A T _E X, high performance computing frameworks Slurm and PBS, GNU Parallel, and MATLAB. Minor experience with Docker and Ansible. |

References

Available on request